A Four-by-Four Obstruction to Polynomial Ordered Binary Matrix Removal

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Candidate research manuscript. Complete written argument and exact finite checks; independent mathematical review, end-to-end formalization, and conclusive novelty assessment remain pending. Developed with Codex assistance.

Abstract

We propose an explicit fixed **4×4 binary pattern** with no polynomial ordered induced removal bound, refining OpenAI's 66×66 obstruction and our earlier 15×15 and 5×5 candidates. The pattern uses a **2×2 anchor component**. The proof classifies embeddings of the complete pattern, including noncanonical anchor configurations, and localizes a fixed zero entry of each copy to a set of only $2m$ host cells.

For every integer $h \geq 1$, put $m = 2^h$. The host has order $(14h+2)m$ and normalized repair distance at least $(14h+2)^{-2}$. Its exact copy count is 128 at $h = 1$, and $\frac{33}{4}m^6 - \frac{29}{2}m^5$ for $h \geq 2$. The resulting density is at most $\frac{33}{4}(14h+2)^{-8}4^{-h}$. Copies match both binary labels and preserve independent row and column orders; repairs allow arbitrary binary entry changes. The fixed pattern works for every pair of positive constants in a proposed polynomial removal bound. Finite checks support the argument but do not establish its universal claim.

Source and contribution

The tree host, four-step propagation mechanism, and sampling argument against arbitrary repairs adapt OpenAI, *Polynomial removal fails for ordered binary matrices*, September 25, 2026, at commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. The qualitative failure of polynomial removal and the 4^{-h} density-decay mechanism are inherited. The proposed refinement is the smaller fixed forbidden pattern, using full-pattern classification, zero-cell localization, and an exact count that includes the additional noncanonical copies.

See `BASELINE.md` and `ATTRIBUTION.md` for source and theorem comparisons. The earlier 5×5 and 15×15 proof drafts and exact reports remain available. The original 66×66 formalization has not been rebuilt here and does not certify these new patterns. No global minimality claim is made.

Statement and explicit pattern

Let

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} S & e_1 & e_2 \\ e_1^T & 1 & 0 \\ e_2^T & 1 & 1 \end{pmatrix}, \qquad P = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

H is the complete forbidden pattern; S is its two-by-two anchor component. H has four distinct rows and four distinct columns, although S's rows and columns are themselves unit vectors. That distinction matters below.

For every integer $h \geq 1$, put $m = 2^h$, $d = 14h + 2$, and $n = dm$. The explicit binary host A_h below satisfies

$$\operatorname{dist}_H(A_h) \ge d^{-2}, \qquad \frac{N_H(A_h)}{n^8} \le \frac{33}{4} d^{-8} 4^{-h}.$$

For $h = 1$ its exact copy count is 128. For $h \ge 2$ it is

$$N_H(A_h) = \frac{33}{4} m^6 - \frac{29}{2} m^5.$$

Copies preserve independently increasing row and column orders and match both binary values. Distance permits arbitrary changes in both directions, including anchor cells, and is normalized by n^2. Consequently for every $c, C > 0$ there are n, $0 < \epsilon < 1$ and a binary A with distance at least ϵ from H-freeness but fewer than $c\epsilon^C n^8$ ordered H copies. The same fixed four-by-four H works for all c,C.

1. Mode order and anchor size

Use $s = 2$ anchor groups per mode on each axis. Order modes on both axes by increasing depth and then

$$W_{i,0}^+,\ W_{i,1}^+,\ W_{i,0}^-,\ W_{i,1}^-,\ V_i^+,\ V_i^-.$$

All anchor positions precede all nonanchor positions. The W-before-V order preserves the containing-supports-come-later property. The order of the two minus parity modes is also used in the new localization argument.

2. Host definition

At depth i of a full binary tree label nodes $p = 0, \dots, 2^i - 1$. At every depth $i < h$ introduce row blocks $R_i^-(p)$, $R_i^+(p)$ and column blocks $C_i^-(p)$, $C_i^+(p)$, each of size $m/2^i$. At depth h there is a single row and column position per node, given both sign names:

$$R_h^-(p) = R_h^+(p), \qquad C_h^-(p) = C_h^+(p).$$

All other blocks on an axis are disjoint. R_i^σ and C_i^σ denote unions over nodes at a depth; each has size m. For $i \ge 1$ the subscript (i,d) restricts to nodes with p congruent to d modulo 2. There are also dummy blocks R_*, C_* of size m.

Order variable rows recursively: a node's minus block, its child-0 subtree, its child-1 subtree, then its plus block. At a leaf insert the one block once. Order columns similarly, with plus before and minus after the subtrees. Put the dummy rows after variable rows, and dummy columns before variable columns.

For every $i = 1, \dots, h$ introduce these six modes:

Mode	Row role 1	Row role 2	Column role 1	Column role 2
V_i^+	R_i^+	R_{i-1}^+	C_*	C_{i-1}^+
V_i^-	R_{i-1}^-	R_i^-	C_*	C_{i-1}^-
$W_{i,0}^+$ and $W_{i,1}^+$	$R_{i,d}^+$	R_*	C_{i-1}^+	$C_{i,d}^+$
$W_{i,0}^-$ and $W_{i,1}^-$	$R_{i,d}^-$	R_*	$C_{i,d}^-$	C_{i-1}^-

Place all mode anchor blocks before all nonanchor positions on each axis. Within an anchor block put s consecutive groups of size m. Use increasing i, then $W_0^+, W_1^+, W_0^-, W_1^-, V^+, V^-$ to

order modes. Each mode's two roles on a given axis are disjoint, even at leaves.

Define every host entry as follows:

- Between anchor groups in the same mode put the corresponding S entry; between different modes put zero.
- A non-anchor position in role j of mode t has the unit-vector signature e_j against that mode's anchor groups. A position in neither role has the zero signature. Apply this independently on both axes.
- A non-anchor-to-non-anchor entry touching a dummy position is one.
- Variable-to-variable entries are zero except for these rules:

Row	Column	Range	Entry
$R_i^+(p)$	$C_i^+(q)$	$0 \le i \le h$	$\mathbf{1}[p \le q]$
$R_i^-(p)$	$C_i^-(q)$	$0 \le i < h$	$\mathbf{1}[p < q]$
$R_i^+(p)$	$C_{i-1}^+(q)$	$1 \le i \le h$	$\mathbf{1}[\lfloor p/2 \rfloor \le q]$
$R_i^-(p)$	$C_{i-1}^-(q)$	$1 \le i \le h$	$\mathbf{1}[\lfloor p/2 \rfloor < q]$

The common leaves satisfy the first rule. The second excludes them; in the final two rules the column signs are still distinct. Hence these assignments are consistent. There are 6hs anchor classes, 2h+1 variable classes and one dummy class of size m on each axis, giving $n = (6hs + 2h + 2)m = (14h + 2)m$. No labels are part of the binary alphabet: the block names serve only to describe actual positions and entries.

The recursive orders imply, writing $\pi(p) = \lfloor p/2 \rfloor$,

$$R_i^+(p) < R_{i-1}^+(a) \iff \pi(p) \le a, \quad R_{i-1}^-(a) < R_i^-(p) \iff a \le \pi(p),$$
$$C_{i-1}^+(b) < C_i^+(q) \iff b \le \pi(q), \quad C_i^-(q) < C_{i-1}^-(b) \iff \pi(q) \le b.$$

This follows because the subtrees at a fixed depth occur as consecutive intervals in increasing node order, and each child's block lies between its own parent's two signs. These comparisons include the shared leaves.

3. Inherited host properties

A nonanchor class means one level/sign class (identifying the two leaf signs as one class L) or the dummy class. A half-class restricts a non-root variable class to one parity of its node label. No auxiliary labels are part of the binary alphabet.

Lemma 1 (comparability and crown exclusion). On nonanchor rows, any two columns from the same nonanchor class have comparable one-sets. The analogous statement holds for rows. Moreover, no nonanchor rows and columns induce the three-by-three crown $Q = J_3 - I_3$, even if their orders are disregarded.

Proof. Within a column class, each variable row's entries form a suffix in increasing node label; dummy rows are constant one. Within a row class, each variable column's entries form a prefix, since all the inequalities in Section 2 decrease as the row node increases. Dummy classes are constant. Thus either same-class row or column pairs cannot contain both traces 10 and 01.

A Q copy has those two traces on every column pair, so its three column classes must be distinct. It cannot include a dummy column, whose entries are constantly one. Classify variable columns by level/sign, with the shared leaves represented by L. Put edges between consecutive levels of each

sign, and a terminal triangle on L,(h-1,+),(h-1,-). A non-leaf variable row can be one only in its own and the preceding class; a leaf row can be one only in those three terminal classes.

Every pair of Q columns has a row with ones on that pair and zero on the third. That row cannot be dummy. Hence every pair of the three column classes must be an edge, forcing the unique terminal triangle. Select columns L(q), $C^{+}_{h-1}(b_{+})$, $C^{-}_{h-1}(b_{-})$. A leaf row p gives the trace

$$(\mathbf{1}[p \le q],\ \mathbf{1}[\lfloor p/2 \rfloor \le b_{+}],\ \mathbf{1}[\lfloor p/2 \rfloor < b_{-}]).$$

These traces form a chain as p increases. They cannot supply both 110 and 101. Non-leaf rows have zero in the L coordinate, and dummy rows give 111. A Q copy requires both missing traces, a contradiction. This includes $h = 1$, when the triangle consists of the leaf class and the two root classes. QED.

Lemma 2 (ordered laminar role supports). Fix either axis. Every role support is a whole nonanchor class or one of its two parity halves. Thus any two role supports are disjoint or nested. If one is a proper subset of another, the containing support belongs to a strictly later mode in the order of Section 1. The two supports of one mode are disjoint.

Proof. The mode table gives the following exhaustive possibilities:

Axis	Whole variable-class supports	Half-class supports
Row	V_i on its depth-i child class; V_{i+1} on its depth-i parent class	$W_{i,d}$ on its depth-i child class
Column	V_{i+1} and $W_{i+1,d}$ on their depth-i parent class	$W_{i,d}$ on its depth-i child class

At depth h the sign classes coincide; full and half leaf supports are still the same whole class and its parity halves. Omit modes with indices outside 1,...,h. Dummy supports are the whole dummy class and are disjoint from every variable support. At the root no half-class support occurs.

The only possible strict inclusion is a parity half of one variable class in the whole of that class. On the row axis, its W_i mode precedes both V_i (by the within-level order) and V_{i+1} (by level). On the column axis, its W_i mode precedes V_{i+1} and $W_{i+1,d}$ by level. Equality of supports across modes, including leaf sign identifications, is harmless. The two roles of a single mode lie in different classes: parent/child, variable/ dummy, or dummy/variable. Therefore they are disjoint. QED.

An anchor row in group j has column-role support j on the nonanchor columns. The analogous statement holds for anchor columns and row-role supports. Here j belongs to {1,2}; there is no third anchor group.

We also use these strict column-order comparisons, obtained directly from the recursive traversal. At any nonleaf level j, and with a child at level i,

$$\begin{aligned} C^{\sigma}_{i}(q) < C^{+}_{i-1}(b) &\iff \lfloor q/2 \rfloor < b, \\ C^{-}_{i-1}(b) < C^{\sigma}_{i}(q) &\iff b < \lfloor q/2 \rfloor, \\ C^{+}_{j}(a) < C^{-}_{j}(b) &\iff a \le b, \\ C^{-}_{j}(a) < C^{+}_{j}(b) &\iff a < b. \end{aligned}$$

The first two apply to either child sign, including the shared leaf. Distinct blocks at one level occur in increasing node order. Positions in the same block may differ, but their entries against any fixed opposite position agree.

4. Every H copy has one of two anchor configurations

Lemma 3. If a,b count the selected anchor rows and columns of an ordered H copy, then $(a, b) = (2, 2)$ or $(2, 1)$.

Proof. Selected anchors form prefixes. Because H has distinct rows and columns, no two selected positions come from the same twin block.

If $a, b \geq 2$, the crossed one-entries (1,2),(2,1) in S and the aligned mode orders force the first two rows and columns into the same mode t: their mode labels satisfy $t(r_1) = t(c_2) \geq t(c_1) = t(r_2) \geq t(r_1)$. Each axis uses that mode's two groups once, in group order. Any further selected anchor would be in a later mode and have signature 00 against those opposite anchors, whereas the remaining H rows and columns require 10 or 01. Hence $a = b = 2$.

Suppose $a \leq 1$. If $b = 0$, rows 2,3,4 on columns 1,3,4 are all nonanchor and form $\left(\begin{smallmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right)$, a row permutation of Q, forbidden by Lemma 1. If $b = 1$ or 2, the first anchor column puts nonanchor rows 2,3 into one role support, hence one class. Those rows have traces 01 and 10 on nonanchor columns 3,4, contradicting same-class comparability. If $b \geq 3$, the first and third anchor columns have signatures 11 and 01 on nonanchor rows 2,3. Their supports intersect, so the first properly contains the third, contradicting Lemma 2's mode order. Thus $a \geq 2$.

If $a \geq 3$ and $b \leq 1$, the first and third anchor rows have signatures 11 and 01 on nonanchor columns 2,3; Lemma 2 again gives an earlier proper container, a contradiction. The only remaining case with $b = 0$ is $a = 2$. Then the second anchor row puts nonanchor columns 1,4 in one class, while nonanchor rows 3,4 have traces 10 and 01 on them, contradicting comparability. These cases leave precisely (2,2) and (2,1). QED.

5. Localization to a small set of zero cells

Let T_h consist of the cells with row the leaf p and column any position of $C^-_{h-1}(\lfloor p/2 \rfloor)$. Each parent block has two positions, so $|T_h| = 2m$. All these cells are zero by the strict minus transition rule.

Lemma 4. Every ordered H copy sends its fixed cell (3,4) into T_h. Its last row is dummy.

Balanced case $a = b = 2$. The remaining selected positions are nonanchor, so their unit signatures force the two designated roles in mode t. The body must be P. Write $\pi(p) = \lfloor p/2 \rfloor$. The original four-case localization applies:

- V^+_i: row order gives $\pi(p) \leq a$; the bottom-right one gives $a \leq b$, forcing the top-right entry to be one, contrary to P.
- V^-_i: row order gives $\pi(p) \geq a$; the top-right zero gives $a \geq b$, forcing the bottom-right entry to be zero, contrary to P.
- $W^+_{i,d}$: column order gives $b \leq \pi(q)$; the top-left one gives $\pi(p) \leq b$. Equal child parity forces $p \leq q$, contradicting the top-right zero.
- $W^-_{i,d}$: column order gives $\pi(q) \leq b$ and the top-right zero gives $\pi(p) \geq b$. Equal parity forces $p \geq q$. Below the leaves the top-left entry $\mathbf{1}[p < q]$ is zero. At the leaves it is $\mathbf{1}[p \leq q]$, forcing $p = q$ and $b = \pi(p)$.

Thus the mode is $W^-_{h,d}$, row 3 is leaf p, column 4 is its parent-minus block, and row 4 is dummy. There are exactly $2m^6$ such H copies.

Unbalanced case $a = 2, b = 1$. Let B be the mode of the second anchor row and first anchor column; their intersection is one, so their groups are opposite. If the row group were 1 and column group 2, then row 3 would have row-role 2 and column 4 would have column-role 1 in B. In V modes the latter is dummy; in W modes the former is dummy. Either way their entry would be

one, contradicting $H(3,4)=0$. Consequently the second anchor row has group 2, the first column has group 1, and

$$r_3 \in \mathrm{rowrole}_1(B), \qquad c_4 \in \mathrm{colrole}_2(B), \qquad A_h(r_3,c_4)=0.$$

Let A be the mode of the first anchor row. Its role support contains both c_2 and c_3, so these columns belong to one class K. The entries against r_3 are 0 and 1, so K is not dummy. The mode order gives $A \le B$. We now treat each possible B; all selected columns satisfy $c_3 < c_4$.

- $B = V_i^+$. Write $r_3 = R_i^+(p)$, $c_4 = C_{i-1}^+(b)$. The zero gives $\pi(p) > b$. If K is the parent-plus class, the one at c_3 gives $\pi(p) \le a$ with $a \le b$ by column order. If K is the own class, the one gives $p \le q$ while $c_3 < c_4$ gives $\pi(q) < b$. Both contradict $\pi(p) > b$. At leaves the only additional class is parent-minus: its one gives $\pi(p) < a$ and its preceding the parent-plus column gives $a < b$, also a contradiction.
- $B = V_i^-$. Write $r_3 = R_{i-1}^-(p)$, $c_4 = C_{i-1}^-(b)$. The zero gives $p \ge b$. In the own class a one at c_3 requires $p < a \le b$, impossible. In the preceding minus class it requires $\pi(p) < a$; column order requires $a < \pi(b)$, again impossible. The preceding class is absent when $i = 1$.
- $B = W_{i,d}^+$. Here $r_3 = R_{i,d}^+(p)$, $c_4 = C_{i,d}^+(q)$, and the zero gives $p > q$. In the own class a one at c_3 requires $p \le q' \le q$. In the parent-plus class it requires $\pi(p) \le a \le \pi(q)$; equal parity of p,q makes this impossible. At leaves a parent-minus c_3 would require $\pi(p) < a < \pi(q)$, also impossible.
- $B = W_{i,d}^-$. Write $r_3 = R_{i,d}^-(p)$, $c_4 = C_{i-1}^-(b)$, so $\pi(p) \ge b$. The parent-minus class cannot contain c_3: a one there would give $\pi(p) < a \le b$. In the own class, column order gives $\pi(q) \le b$. If $i < h$, the one requires $p < q$. Because $A \le B$, a role support for this own minus class can only be a child half-class of $W_{i,e}^-$ with $e \le d$. Whole-class supports occur at level i+1. Thus q has parity $e \le d$, and $\pi(p) \ge b \ge \pi(q)$ forces $p \ge q$, a contradiction. Therefore $i = h$. In the leaf class the one instead gives $p \le q$, hence $\pi(p) \le \pi(q) \le b \le \pi(p)$, so $b = \pi(p)$. The only remaining possibility is a parent-plus c_3. Its one gives $\pi(p) \le a$, while its position before the parent-minus c_4 gives $a \le b$. Again $b = \pi(p)$.

This proves localization in both cases. In the unbalanced case K is either the leaf class or the parent-plus class. If K is the leaf class, c_2 is a leaf $q_1 < p$ because $A_h(r_3,c_2)=0$. A nonanchor row one at c_2 must be a leaf of index at most q_1, or dummy; a later row r_4 cannot be such a leaf. If K is parent-plus, c_2 has parent node $a < \pi(p)$. A nondummy row one at both c_2 and the parent-minus c_4 must be a leaf. Its being one at c_2 gives parent node at most $a < \pi(p)$, incompatible with its occurring after leaf p. Hence r_4 is dummy in both cases. QED.

In particular, the old leaf-diagonal hitting set is insufficient: unbalanced copies can have no leaf column at all. The new fixed-cell localization is to zeros, and remains useful because induced copies must match zeros too.

Localization alone already gives $N_H(A_h) \le |T_h| n^6 = 2mn^6$, hence $N_H(A_h)/n^8 \le 2d^{-2}2^{-h}$. Together with the distance bound below, this suffices to refute every polynomial removal bound. The exact enumeration in the next section strengthens the density estimate; the qualitative obstruction does not depend on that enumeration.

6. Exact copy count, including the additional copies

The balanced count is $2m^6$: m leaf choices, m dummy row choices, two parent-column positions, and four independent size-m anchor choices.

For an unbalanced copy, fix its leaf $p = 2b + d$, with $d \in \{0, 1\}$. Its second anchor row and first anchor column have mode $B = W^-_{h,d}$ and groups 2,1. Its last column has two positions in the parent-minus block b, and its last row has m dummy choices. The three anchors contribute m^3. Thus each pair of actual positions (c_2, c_3), together with the first anchor's mode and group, contributes $2m^4$ copies. The following list is exhaustive:

Class K	First anchor mode/group	Allowed pair count
Leaf	$W^+_{h,e}$, group 2, $e \geq d$	b for each e
Leaf	$W^-_{h,d}$, group 1	b
Parent-plus	$W^+_{h,e}$, group 1, either e	4b for each e
Parent-plus	$W^+_{h-1,e}$, group 2, $e = b \bmod 2$, if $h \geq 2$	$4\lfloor b/2 \rfloor$

For the leaf class, c_3 is the unique leaf 2b+e, while c_2 can be any earlier leaf of parity e below p, giving b choices. The minus mode order and the required one imply $e = d$ in the minus case. For parent-plus columns, c_3 has node b and c_2 has node $a < b$; each parent block has two actual positions, so a node pair has multiplicity four. Half-class supports restrict a and b to the same parity. The remaining possible full parent-plus support belongs to V^+_h, later than B, so cannot occur. Whole leaf supports do not exist on the column axis within the available levels.

Conversely every pair listed gives the displayed H, in order: the first anchor mode is earlier than B (or is B with group 1), all required role signatures hold, the dummy row gives its three nonanchor ones, and the leaf thresholds give entries 0,1,0 on columns 2,3,4. Thus no overcount or missing family is hidden in the enumeration.

For $p = 2b + d$ the total pair count is $(11 - d)b + 4\lfloor b/2 \rfloor$. At $h = 1$, $b = 0$ for both leaves, so there are no unbalanced copies. For $h \geq 2$ let $t = m/2$, which is even. Summing both parities,

$$\begin{aligned}\sum_{b=0}^{t-1}(21b + 8\lfloor b/2 \rfloor) &= \frac{21t(t-1)}{2} + 2t(t-2) \\ &= \frac{t(25t - 29)}{2}.\end{aligned}$$

Multiplication by $2m^4$, then addition of $2m^6$, gives $N_H = \frac{33}{4}m^6 - \frac{29}{2}m^5$ for $h \geq 2$. At $h = 1$, $N_H = 128$. In every case $N_H \leq \frac{33}{4}m^6$.

7. Distance against arbitrary binary repairs

Let B differ from A_h on a set E of $M < m^2$ cells. Partition each axis into the size-m anchor groups, dummy group, the two sign classes at each depth below h, and the single shared leaf class.

Choose a uniform leaf z in {0,. . . ,m-1} and its ancestor $a_i = \lfloor z/2^{h-i} \rfloor$. Choose one uniform position x^σ_i, y^σ_i from the row and column blocks with that ancestor at each level and sign. At the leaf use one common position for both signs on each axis. Conditional on z, make all other choices independently. Also choose independently one uniform representative of every anchor and dummy group on each axis.

Each variable position has marginal sampling probability $2^{-i}/(m/2^i) = 1/m$. Anchor and dummy selections are independent of every opposite-axis selection. The root selections, whose node is always 0, are independent uniform choices too. Therefore every cell with an anchor or dummy endpoint, and every root-to-root cell, is selected with probability exactly $1/m^2$. Within

each class there is only one selection. A union bound over edited cells of these types gives probability at most $M/m^2 < 1$ that any such selected cell was edited. Fix a selection with none edited.

This statement does NOT apply to arbitrary variable-to-variable cells: for example a fixed diagonal leaf cell is selected with probability 1/m, because the two axes share z. No such independence is used below.

This argument uses only the designated mode embeddings; it does not require all copies of S to be rigid. Suppose B were H-free. Its selected anchors still form S, and their signatures against selected body positions and all dummy incidences are unchanged. Thus whenever a mode's selected body is P, it extends to H in B. Its absence forces the following implications.

The unchanged root-plus diagonal is one and root-minus diagonal is zero. Suppose at depth i-1 these values are still one and zero. The V_i^+ and V_i^- bodies respectively have form

$$\begin{pmatrix} 1 & b_+ \\ 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 1 & b_- \end{pmatrix}.$$

Avoiding P forces $b_+ = 1$ and $b_- = 0$. These are the entries from selected depth-i rows to selected depth-(i-1) columns. Next use the W modes with $d = a_i \bmod 2$. Their bodies respectively have form

$$\begin{pmatrix} 1 & c_+ \\ 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} c_- & 0 \\ 1 & 1 \end{pmatrix}.$$

Avoiding P forces $c_+ = 1$ and $c_- = 0$ at the depth-i diagonals. All four bodies are in increasing row/column order because their selected nodes are parent and child, and dummy positions have the stipulated extreme orders. Induction carries opposite values to depth h. At that depth the plus and minus diagonals are the same cell, a contradiction. Hence B is not H-free.

Every H-free repair requires at least m^2 changes. Dividing by $n^2 = d^2m^2$ gives $\mathrm{dist}_H(A_h) \geq d^{-2}$. No restrictions were imposed on unprotected entries or on the directions of the changes.

For a concrete distinction between hitting and repairing, flipping every cell of T_h from zero to one destroys all original copies but creates new ones. In mode V_h^- choose body rows $R_{h-1}^-(b)$, $R_h^-(p)$ with $\pi(p) = b$, and body columns C_*, $C_{h-1}^-(b)$. Their entries after these flips are P: the parent-minus diagonal remains zero and the leaf-to-parent entry is now one. The unchanged designated anchors extend this body to H.

8. Final implication and sampling consequence

For $\epsilon_h = (14h+2)^{-2}$, the exact count gives, for every fixed $C > 0$,

$$\frac{N_H(A_h)}{\epsilon_h^C n_h^8} \leq \frac{33}{4}(14h+2)^{2C-8}4^{-h} \longrightarrow 0.$$

Choosing h sufficiently large for any prescribed $c > 0$ proves the announced failure of every polynomial removal bound, with H fixed first.

For canonical sampling of q uniform rows and independently q uniform columns, observing H requires sampling a cell in T_h. Therefore

$$\Pr(\text{reject}) \leq 2m(q/n)^2 = 2\epsilon_h 2^{-h}q^2.$$

Rejection probability at least $\rho > 0$ requires

$$q \geq \sqrt{\rho/2}\,(14h+2)2^{h/2} = \sqrt{\rho/2}\,\epsilon_h^{-1/2}2^{(\epsilon_h^{-1/2}-2)/28}.$$

There are q^2 observed entries. This is a canonical sampler statement, not a claim about all adaptive or two-sided testers. The dependence remains $\exp(\Omega(\epsilon^{-1/2}))$; the new result is the smaller forbidden pattern.

Verification boundary and comparison

The pinned primary obstruction is 66×66. The earlier local candidates were 15×15 and 5×5, with anchors of size 13 and 3. This candidate uses a 4×4 pattern with a 2×2 anchor component, and allows genuine unintended anchor copies. The previous limitation on pure two-by-two anchor rigidity remains true; it does not preclude this whole-pattern localization argument.

check_four.py exhaustively checks the full pattern through depth 4 without assuming localization, including exact weights of twin blocks. A separate node-pair summation checks the closed count through depth 8. The checker also tests the required tree orders, nonanchor crown exclusion, laminar role supports, protected sampling probabilities, and arbitrary path assignments. The all-depth proof above is separate from finite evidence. No independent review or Lean kernel verification is claimed; novelty screening remains inconclusive.

check_four_repair.py additionally checks the explicit new copies produced by flipping T_h, for every leaf through depth 8.